Requirements Engineering for Agentic AI systems: sustainability, transparency and observability

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Abstract. Agentic AI represents a paradigm shift from reactive artificial intelligence systems to autonomous, adaptive, and goal-oriented agents, with far reaching economic, societal and environmental impacts. This paper outlines a research proposal, trying to answer the question: can methods, models and artefacts of Requirements Engineering be adapted and extended to specify, implement and evaluate, simultaneously, sustainability, transparency and observability requirements in Agentic AI? Expected contributions are outlined, together with achieved results and an evaluation plan for the stated contributions. The major result achieved so far is a list of best practices to write AI agents, through requirements of three main classes: sustainability, transparency and observability.

Keywords: Agentic AI, Requirements Engineering, Sustainability, Transparency, Observability.

1 Introduction

Since 2022 there has been an increasing adoption of artificial intelligence (AI) by both individuals and organizations [3]. Among them we can find Generative AI tools, e.g. ChatGPT, and Agentic AI tools, e.g. CrewAI. Agentic AI adds autonomy, planning and reasoning capabilities to AI, with increasing impacts across the economy, for example, through higher productivity, in finance, robotics, healthcare [10]. Nevertheless, fears regarding increasing unemployment, training of human resources and contributions to climate change and environmental degradation are also a reality [7][10]. In this context, the contribution of Requirements Engineering (RE) can be possibly valuable to boost the positive impacts of Agentic AI, increasing efficacy and efficiency, through the elicitation of requirements, for three classes of requirements, namely: sustainability, transparency and observability. In other words, the core problem of this research is: can methods, models and artefacts of Requirements Engineering be adapted and extended to specify, implement and evaluate, simultaneously, sustainability, transparency and observability requirements in Agentic AI? It reflects concerns at the age of AI [27].

In fact, several authors mention the need for a unified model of RE for Agentic AI to allow its contribution to sustainable development, i.e. development that satisfies present needs without compromising the needs of future generations [2][22][33]. This means that clear metrics are needed to support Ethical Governance [29]. Nevertheless,

because traditional RE only looks to static requirements it is not obvious how this contribution of RE will take place, due to the autonomous and dynamism of Agentic AI.

The structure of this paper is as follows: (Section 2) Related Work; (Section 3) Expected Contributions; (Section 4) Methodology; (Section 5) Results Achieved; (Section 6) The Evaluation Plan; and (Section 7) Conclusion.

2 Related Work

Agentic AI is a new concept of AI that has capabilities of memory, reasoning, autonomy, planning and interaction with external tools, using large language models (LLMs), i.e., automatic learning systems created to predict and generate plausible language [2]. In this research, it implies the usage of AI agents, i.e. software entities with capabilities of reactivity, persistence, proactivity, autonomy, reasoning and planning, corresponding to Multi-Agent Systems and increased efficacy when dealing with complex problems [2-3][29]. Figure 1 shows the functioning of Agentic AI.

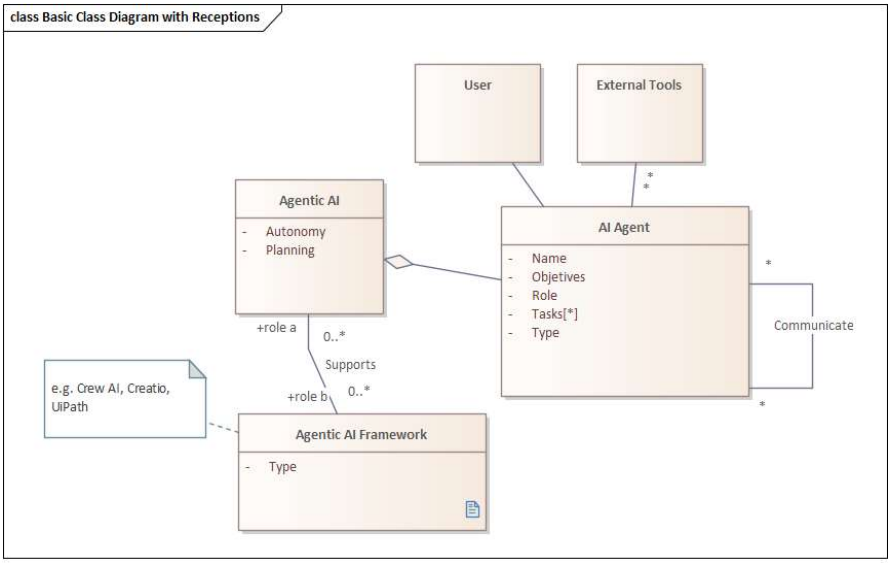


Fig. 1. Functioning of Agentic AI

2.1 Challenges in Requirements Engineering and Agentic AI

Nevertheless, the challenges for RE in Agentic AI are the ones from AI agents, such as hallucination, limited planning ability, prompt brittleness, and lack of causal understanding [29]. Additionally, for Agentic AI, there are challenges as inter-agent misalignment, emergent behavior, error propagation, adversarial vulnerabilities and explainability deficits [29]. In fact, RE implies the identification of functional and quality requirements since the very first steps of software development [26]. Furthermore, different domains imply distinct sets of requirements. For example, in healthcare Hadar, E., Hadar, I., Levy, M. and Zlotnik, S. (2025) [11] use RE to create a digital mentor and

identify classes of requirements such as trustworthiness and explainability. In education and science, Jaradeh and Auer (2025) [14], stress the planning capacity, for long periods of time. Sarferaz [30] stress the specialization of agents to deal with problem complexity. Also, common to all domains, in ethics, there is a growing preoccupation with climate change and environmental degradation [7]. The use of these requirements in defining AI agents has been analyzed by Schneider [31], who highlights the role of prompt engineering, understood as the creation of natural language instructions to structure dynamic behaviors, hierarchical relationships, and execution cycles, enabling LLMs to plan actions, and use tools and feedback mechanisms.

At the same time, the MCP protocol [21], which is an open standard, allows programmers to set up secure two-way connections between data sources and AI tools. Additionally, the Agent-to-Agent (A2A) protocol [1] defines mechanisms that make it easier for agents and different platforms to work together. Regardless of the availability of these resources and knowledge, because there is no simple and holistic standard model to study Agentic AI, integrating social, economic and environmental dimensions [2][22], and RE faces difficulties to provide valuable holistic metrics and solutions for higher levels of sustainability, observability and explainability. Therefore, Sections 2.2 to 2.4 indicate requirements identified during this research that might support a new model for Ethical Governance frameworks as mentioned by Sapkota et al. [29].

2.2 Sustainability Requirements

The sustainability of software products, including AI agencies, is the efficient use of energy and water, promoting positive impacts on socioeconomic activity, and minimizing the negative environmental impacts associated with the processes that support their development and operation [34]. This perspective, focused on the relationship between humans and the planet, integrates the environmental, social, and economic dimensions of sustainability [32][34]. The KarlsKrona Manifesto [4] and the concept of Green in IT [24], one focus of this research, referring to the development of more environmentally sustainable software products, help the operationalization of sustainability requirements. Additionally, Christadoss and Panda [7] identify as a central question the assessment of the balance between CO_2 emissions produced by the technology and the emission reductions its use can provide.

In an AI agency, for example, the following sustainability requirement is suggested: Shall calculate electricity consumption per agent or per prompt to assess the possible CO_2 emissions to reduce [7]. Additionally, generic requirements like the ones that will be presented in Section 5, can contribute to the accountability of AI agents and the quality of the results obtained, becoming a step toward responsible use of resources, that is, efficient, effective, and ethical. But despite its importance sustainability requirements rely on explainability and observability requirements to support their rational and guarantee the acceptance of the supposed environmental aware solutions.

2.3 Observability Requirements

Observability is defined as the ability to observe, monitor and understand, in real time, the system behavior [20]. It is a concept closely linked to governance, that in Agentic

AI aims to balance two potentially conflicting goals [20][24]: the autonomy and adaptability of agents, on one hand, and organizational supervision and control, on the other, requiring observable technical data to achieve them, such as events and resource usage. In this context, Patil [24] proposes a governance platform that integrates: (i) requirements arising from new legal frameworks, including European regulations like the EU AI Act [9]; and (ii) emerging technical solutions. Despite these advances, significant limitations persist. Joshi et al. (2025) [15] identify gaps associated with regulatory fragmentation across different jurisdictions and the predominantly static nature of existing governance mechanisms and Schneider [31] highlights the use of structured sequences of prompts to regulate SMA behavior, an approach called Agent Orchestration, through which it is possible to define operational limits, requirements, objectives, and agent identities, i.e. the dynamic coordination [29].

The following observability requirements are suggested [9][20]: (i) Shall compute the values of the memory and CPU resources used by the agents; (ii) Shall compute parameters like temperature, top-k of the LLMs used; (iii) Shall compute logs, i.e., chronological records, of the tools used by the agents; (iv) it Shall not score citizens.

2.4 Transparency or Explainability Requirements

Transparency in an AI agency involves not just explaining the system's behavior and decisions but also showing whether it acts fairly and in line with ethical and legal principles for all stakeholders. Transparency can be measured by defining and evaluating specific requirements, contributing to trustworthiness and making it easier to identify and reduce systematic biases, that is, systematic deviations from the social values being defended [28][35]. The following transparency requirements are suggested [29]: (i) Shall provide shared logs, e.g. record memory updates, tool calls, record prompts; (ii) Shall provide interpretable reasoning paths across agents; (iii) Shall provide response verification through validation protocols; and (iv) Shall explain the obtained results.

The data gathered for observability is mainly for teams of software developers, but the data gathered for transparency and explainability should be understandable by all stakeholders. Having this in mind in the next sections the research questions and the evaluation plan are further detailed.

3 Expected Contributions

The expected contributions try to answer two Research Questions (RQs), namely:
(RQ1) What are the relevant requirements to define and evaluate sustainability, transparency and observability of Agentic AI?
(RQ2) What are the best practices of RE to promote sustainability, transparency and observability of Agentic AI?
Expected contributions for RQ1: RQ1 is like a roadmap for the identification of requirements to be elicited and classified according to three main classes of requirements named sustainability, transparency and observability. The contribution (C1) is the

identification of requirements that improve the performance and acceptance of Agentic AI, through Tasks 1, 4 and 5 in Section 6, implying new simulations.
Expected contributions for RQ2: (C2) the identification of a multi-criteria metrics using Agentic AI requirements to holistically evaluate the environmental impact of Agentic AI (i.e. it sums up the impact of all the relevant distinct variables in a single value, simplifying the study of the software complexity [19]); (C3) to propose best practices and a model for Agentic AI (i.e. a conceptual framework, useful for governance and design), integrating sustainability, transparency and observability. (C2) through Tasks 2, 4 and 5 in Section 6. (C3) through Tasks 3, 4 and 5 in Section 6.

4 Methodology

The adopted methodology is the Design Science Research (DSR) which is an action sequence [12]: (i) identification of research problem and motivation; (ii) identification of objectives for the future solutions; (iii) design solution, i.e. the design and creation of the artifact for the problem; (iv) demonstration, through tests and simulations of the new tools, metrics and prompts; (v) evaluation, i.e. the assessment if the proposed solutions, for example, using usability questionnaires [6], with a target satisfaction of 80 %, and expert interviews; (vi) communication; and (vii) iteration according to the evaluation. The demonstration uses simulations [5], as further detailed in Section 6.

The expected solution, i.e. in step (iii), should result from the expected contributions, as an example of more sustainable usage of Agentic AI, e.g. consuming less electricity for the same results, using metrics for more than one purpose, obtaining increased trust.

5 Results Achieved

The main results achieved include the (i) identification of a multi-criteria metrics for environmental sustainability, the identification of (ii) sustainability, (iii) transparency and (iv) observability requirements. Results (i) and (ii) at published papers are available at: https://github.com/reviewpaperq/papers.

Additionally, tests already performed and the analyzed literature allow the identification of several generic recommendations to write AI agents, implemented with requirements, as mentioned in section 2, namely [3][11][13][16-18][23][29][30]:
R1. Before writing an AI agent, you shall understand and identify its description, objectives, requirements, tasks, tools and expected results;
R2. An AI agent shall start with a meaningful identification, that can be structured through a name, description and intent, that might be helpful for error accountability;
R3. The chosen prompting technique shall be adapted to the application domain together with proper safety and legal procedures, e.g. in finance, and healthcare;
R4. The prompt pattern shall be rigorous, using direct speech (as this speech style is informative), and structured in sections, eliciting objectives, tools and requirements;
R5. A possible sequence of sections shall be: name, description, objectives, requirements, tasks, questioning enforcement, hallucination detection, expected results, avail able knowledge, tool identification and environmental sustainability;

R6. The first approach and draft prompt shall be produced with the help of Gen AI tools i.e. through a meta-prompt, and be prepared for AI agents' changing objectives;
R7. There shall be approaches to overcome the limits of LLMs;
For example, LLMs' context windows are too small to handle complete websites, leading to shallow reasoning and hallucinations. This shall be handled with the creation of a llms.txt file with LLMs' guiding content, written in markdown language.
R8. An AI agent shall consider a human-in-the-loop approach, e.g., for evaluations and domain knowledge requirements;
R9. An AI agent shall include guardrails and exception handling, e.g. in industry;
R10. An AI agent shall include mechanisms for hallucinations' detection, e.g. classified as intrinsic or extrinsic, through automatic data validation or human feedback;
R11. An AI agent shall use specialized code to access external online resources;
R12. You shall adopt an iteration approach due to the evolving reality and the positive impacts of training LLMs, using a cycle of test-then-improvements then new tests;
R13. You shall add procedures, such as the test of agents and the size reduction of specification prompts, to increase their efficiency and reduce environmental impacts;
This R13 implies the decreased consumption of electricity due to the avoiding of unnecessary computational instructions.
R14. You shall use retrieval augmented generation (RAG) when outputs need real-time data, e.g. in domains like enterprise research and customer support.

This set of generic best practices is possibly novel in RE terms because it groups together several requirements to allow the creation of effective AI agents, in a less complex way. The three classes of requirements, previously mentioned in Section 2, add to this set implying the calculus of metrics and contributing to a performance that respects social values, e.g. sustainability and law compliance, increasing AI acceptability.

6 The Evaluation Plan

The expected contributions are obtained with tasks depicted at Figure 2, where Sem states for Semester. T1 and T2 continue from previous work due to the evolution of Agentic AI [22].

Task	2026/2027 Sem 1	2026/2027 Sem 2	2027/2028 - 3 months
T1: To identify FR, QR and metrics			
T2: List of good practices			
T3: Model creation			
T4: Validation			
T5: To repeat tasks T1 to T4			
T6: To write the thesis			

Fig. 2. Tasks over time.

The list of best practices is evaluated using an evaluation plan with several steps. First, three main use cases are created: (i) the tourism agent; (ii) the researcher; (iii) the financial investor. Second, prompts for each use case are created, with and without the identified best practices, which are enforced using natural language. Third, simulations

over 200 runs are performed to compare quantitatively and qualitatively the performance of online agentic AI tools, and local LLMs, e.g. LLaMA 3-3. Quantitative metrics are, e.g. latency, electricity consumption (KWh), number of tool calls, hallucinations and multi-criteria metrics; and qualitative metrics are, e.g. completeness. The simulations follow as close as possible examples of good simulation practice [5], all metrics are linked with the requirements identified in Section 2, and results are extracted to separate files, and then computed together being analyzed with regression analysis if needed [8]. Fourth, questionnaires and, if possible, interviews are conducted. Fifth, improvements are made before new evaluations. Finally, the connection between the model and the best practices is explained.

7 Conclusion

This paper presents a research proposal to answer the identified research problem and discusses how to evaluate the main findings. Future research work includes the execution of the proposed evaluation plan and the identification of new challenges and tests.

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